

Full Stack Development and Deployment of a Web-Based Speech Recognition System

Week 18-19
11 May - 24 May 2026

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Project Overview

- MAGOR is a web-based audio and video transcription system that converts recordings into text
- **Existing Problem:**
 - Although functional, components remain incomplete and require refinement
- **Proposed Solution:**
 - Improve robustness, usability, and AI-driven recording analysis



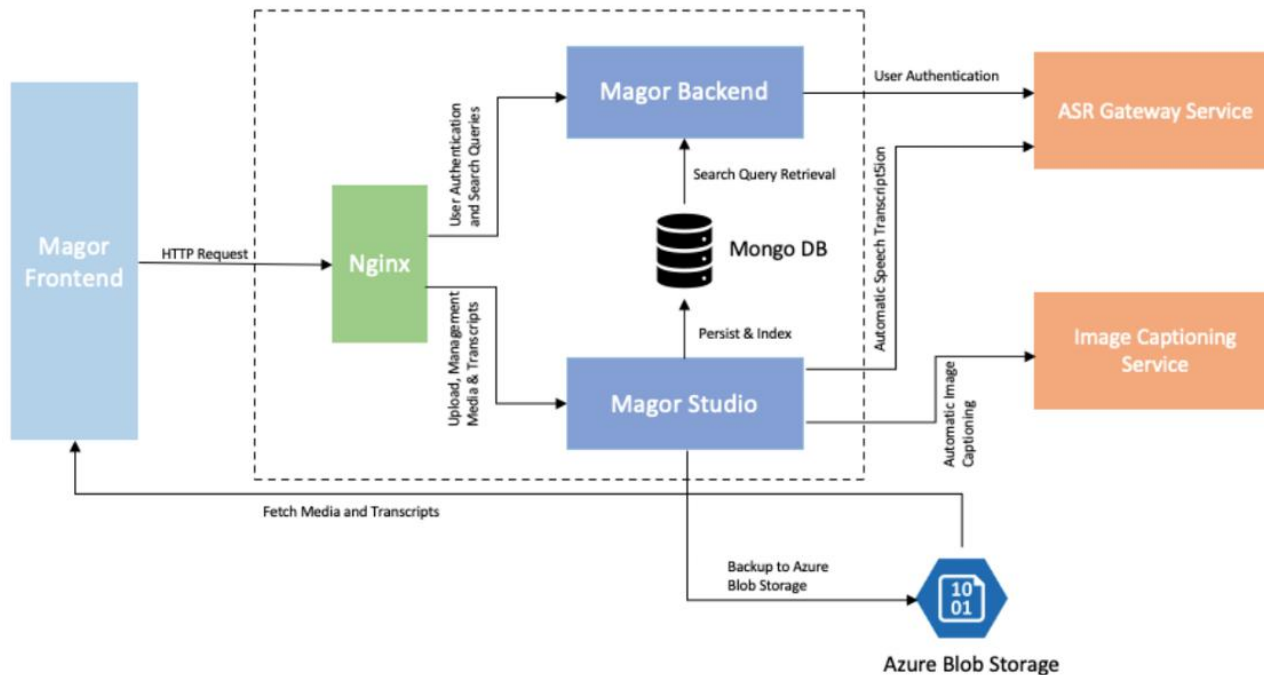
Project Objectives

The primary objectives of this project are:

- Fix system bugs and restore functionality
- Improve usability and user experience
- Enhance maintainability and scalability
- Add AI features to existing pages



Systems Overview



Source: Huang (2021)



Agenda

Work done this week:

1. Data Volumes

- Delete image file upon recording deletion

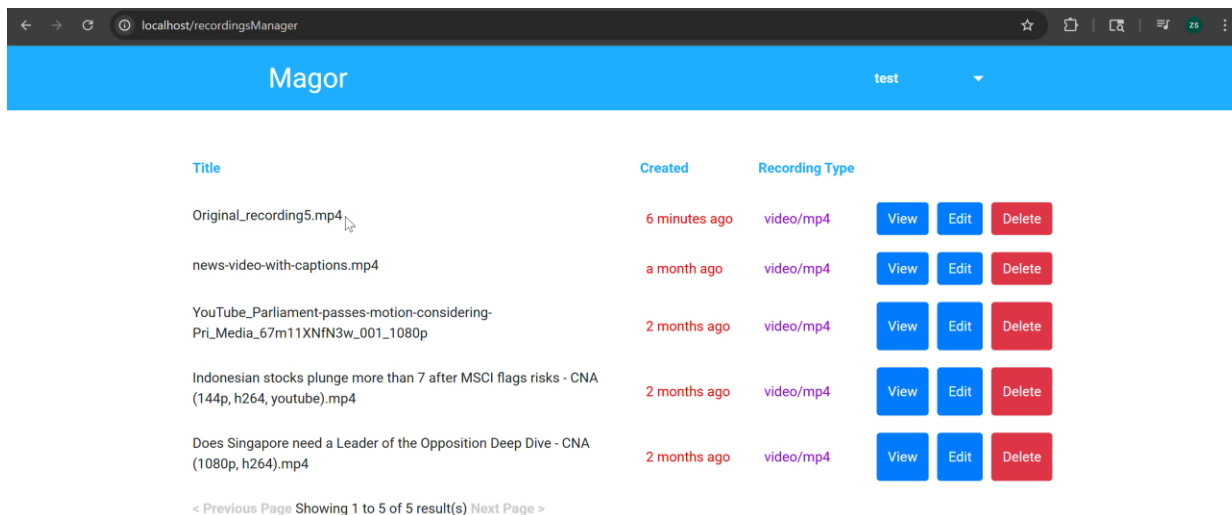
2. Admin ASR Request Page

- Fixed errors when loading ASR requests
- Updated the system to retrieve transcription jobs from the correct source
- Fixed deletion of ASR requests
- Update transcript job status of deleted recordings to “deleted”
- Performed UI updates and functionality verification



Task 1 (Delete Image File Upon Recording Deletion)

- Delete recording and associated files demo:



The screenshot shows a web browser at the address `localhost/recordingsManager`. The application has a blue header with the text "Magor" and a dropdown menu currently showing "test". Below the header is a table with three columns: "Title", "Created", and "Recording Type". The table contains five rows of recording data. Each row has three buttons: "View" (blue), "Edit" (blue), and "Delete" (red). At the bottom of the table, there is a pagination link: "< Previous Page Showing 1 to 5 of 5 result(s) Next Page >".

Title	Created	Recording Type			
Original_recording5.mp4	6 minutes ago	video/mp4	View	Edit	Delete
news-video-with-captions.mp4	a month ago	video/mp4	View	Edit	Delete
YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	2 months ago	video/mp4	View	Edit	Delete
Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	2 months ago	video/mp4	View	Edit	Delete
Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	2 months ago	video/mp4	View	Edit	Delete

< Previous Page Showing 1 to 5 of 5 result(s) Next Page >



Task 2 (Restore Admin ASR Request Page)

- Error loading ASR request demo:

Users

BACK TO MAGOR

recordings

asrRequests

Statistics

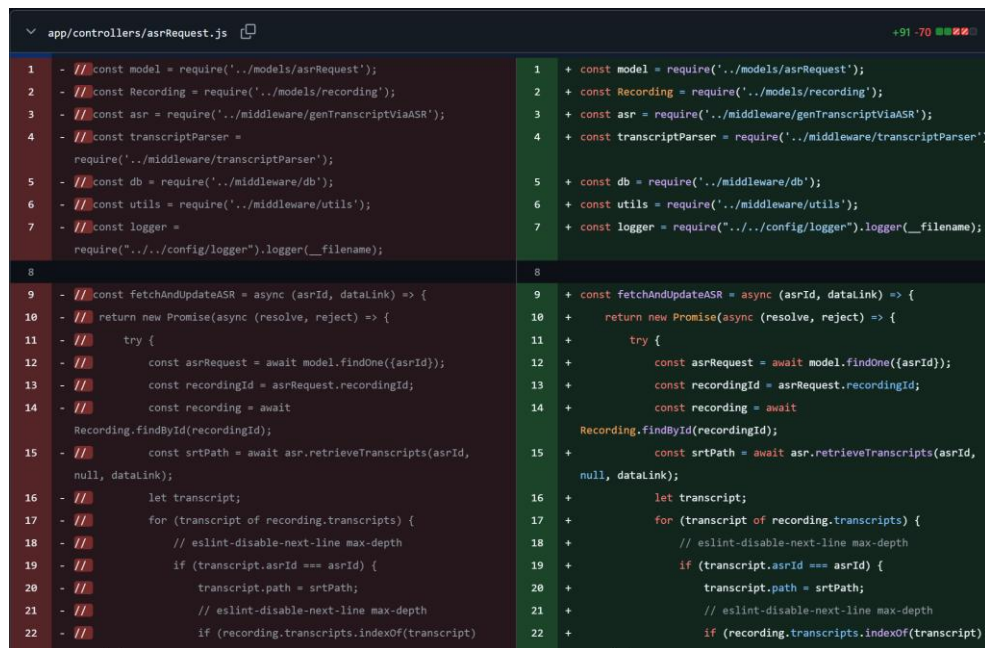
☐	Email	Name	Role	User status		
☐	test@docker.com	test	admin	verified	EDIT	DELETE
☐	testuser1@gmail.com	testuser1	user	unverified	EDIT	DELETE
☐	derick@outlook.com	derick	user	unverified	EDIT	DELETE
☐	newuser@test.com	newUser	user	unverified	EDIT	DELETE
☐	new@test.com	newuser	admin	unverified	EDIT	DELETE
☐	user2@test.com	newuser2	user	unverified	EDIT	DELETE
☐	tee@te.com	tee	user	unverified	EDIT	DELETE
☐	derick@hotmail.com	derick	user	unverified	EDIT	DELETE

Rows per page: 10 1-8 of 8



Task 2 (Restore Admin ASR Request Page)

- Code Changes:
 - Backend code was commented out



```
app/controllers/asrRequest.js

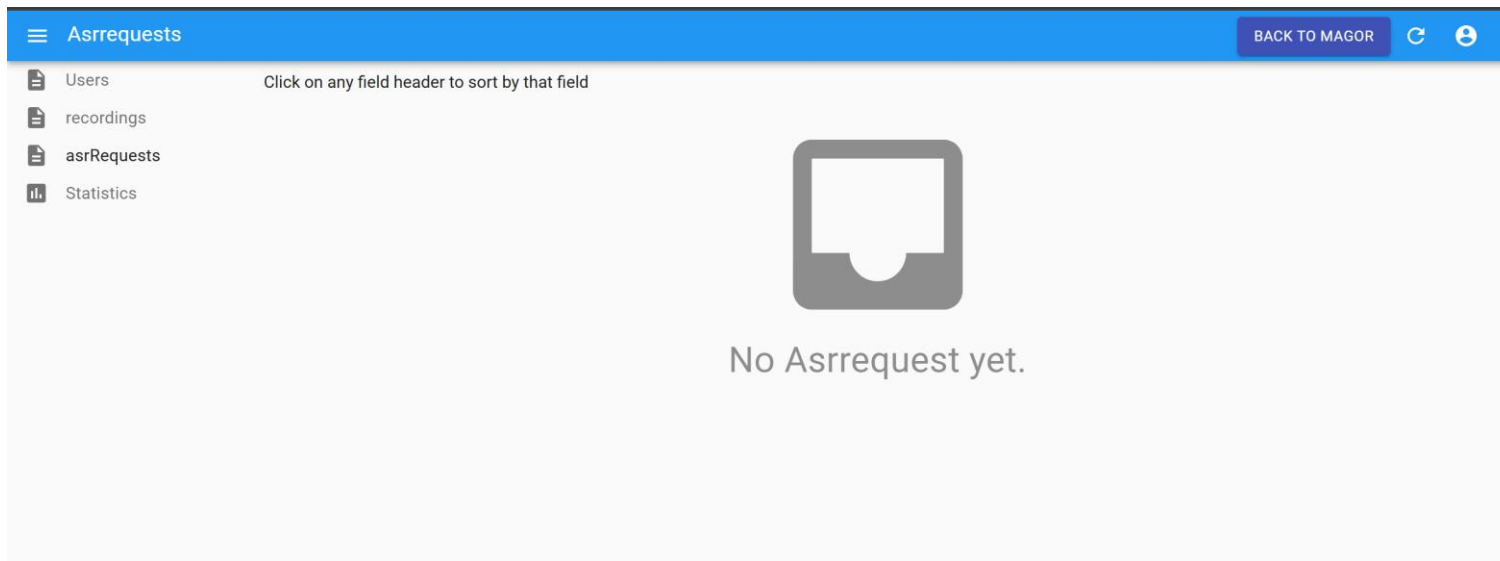
1 - //const model = require('../models/asrRequest');
2 - //const Recording = require('../models/recording');
3 - //const asr = require('../middleware/genTranscriptViaASR');
4 - //const transcriptParser =
  require('../middleware/transcriptParser');
5 - //const db = require('../middleware/db');
6 - //const utils = require('../middleware/utils');
7 - //const logger =
  require("../../config/logger").logger(__filename);
8
9 - //const fetchAndUpdateASR = async (asrId, dataLink) => {
10 - //  return new Promise(async (resolve, reject) => {
11 - //    try {
12 - //      const asrRequest = await model.findOne({asrId});
13 - //      const recordingId = asrRequest.recordingId;
14 - //      const recording = await
  Recording.findById(recordingId);
15 - //      const srtPath = await asr.retrieveTranscripts(asrId,
  null, dataLink);
16 - //      let transcript;
17 - //      for (transcript of recording.transcripts) {
18 - //        // eslint-disable-next-line max-depth
19 - //        if (transcript.asrId === asrId) {
20 - //          transcript.path = srtPath;
21 - //          // eslint-disable-next-line max-depth
22 - //          if (recording.transcripts.indexOf(transcript)

1 + const model = require('../models/asrRequest');
2 + const Recording = require('../models/recording');
3 + const asr = require('../middleware/genTranscriptViaASR');
4 + const transcriptParser = require('../middleware/transcriptParser');
5 + const db = require('../middleware/db');
6 + const utils = require('../middleware/utils');
7 + const logger = require("../../config/logger").logger(__filename);
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  Recording.findById(recordingId);
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  null, dataLink);
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20 +           transcript.path = srtPath;
21 +           // eslint-disable-next-line max-depth
22 +           if (recording.transcripts.indexOf(transcript)
```



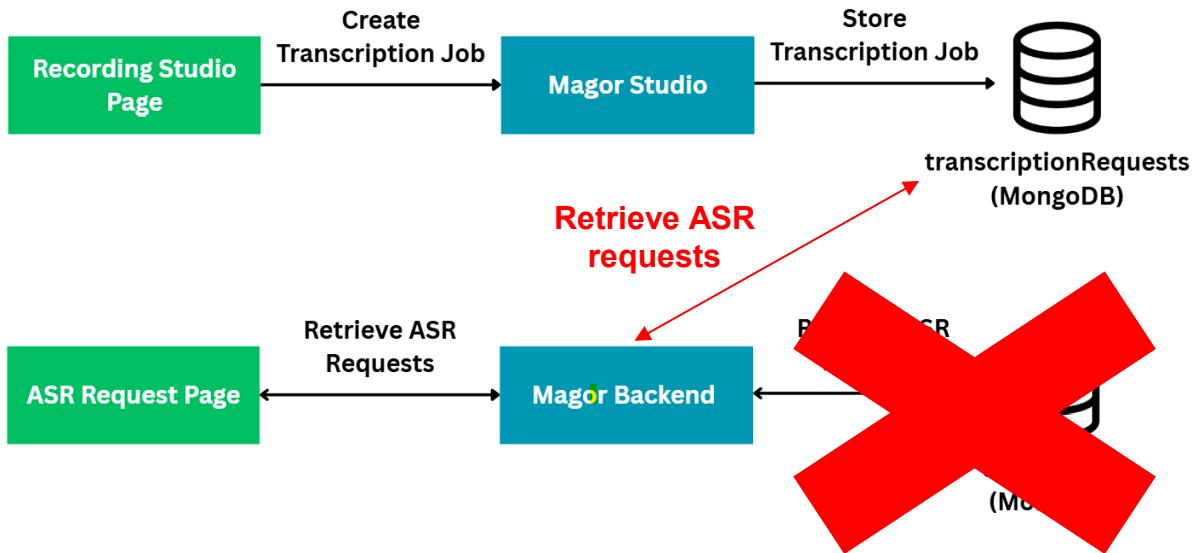
Task 2 (Restore Admin ASR Request Page)

- ASR requests now calls backend:
 - No ASR request retrieved due to incorrect data source handling



Task 2 (Restore Admin ASR Request Page)

- ASR requests now calls backend:
 - No ASR request retrieved due to incorrect data source handling



Task 2 (Restore Admin ASR Request Page)

- ASR requests now calls correct source:

ASR Requests

BACK TO MAGOR

Users

recordings

ASR Requests

Statistics

Click on any field header to sort by that field

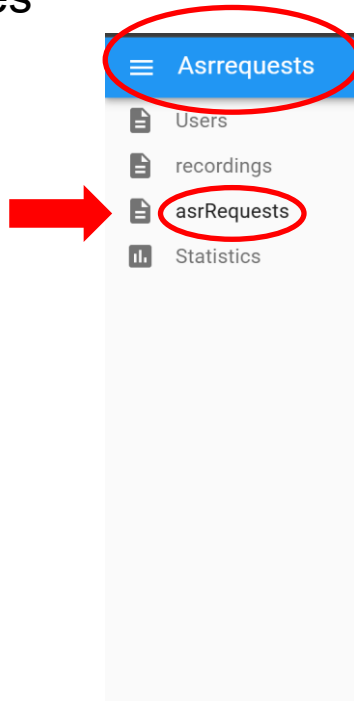
EXPORT

☐	recordingId ↑	Recording ↑	Asr	Status	date	
☐	69afcce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	success	3/10/2026	View on Magor
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	success	3/11/2026	View on Magor
☐	69b2730fdb3f0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	success	3/12/2026	View on Magor
☐	6a14889a49b93b00234a46cd	Original_recording5.mp4	6a1488a263309cdb335cf350	success	5/26/2026	View on Magor

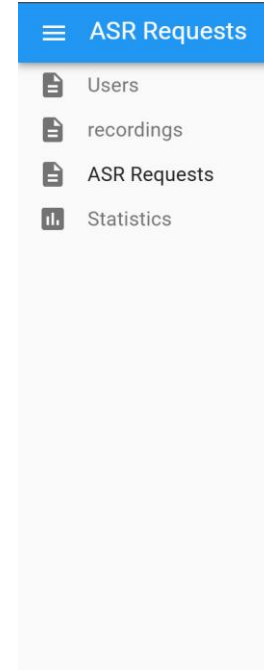
Rows per page: 10 1 of 4

Task 2 (Restore Admin ASR Request Page)

- UI Updates



Before

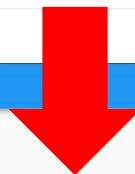


After



Before

ASR Requests							BACK TO MAGOR		
Users	Click on any field header to sort by that field								
recordings									
ASR Requests									EXPORT
Statistics									
☐	recordingId ↑	Recording ↑	Asr	Status	date				
☐	69afce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	success	3/10/2026		View on Magor		
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	success	3/11/2026		View on Magor		
☐	69b2730fdb3f0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	success	3/12/2026		View on Magor		
☐	6a14889a49b93b00234a46cd	Original_recording5.mp4	6a1488a263309cdb335cf350	success	5/26/2026		View on Magor		
							Rows per page: 10	1-4 of 4	



Heading Update

After

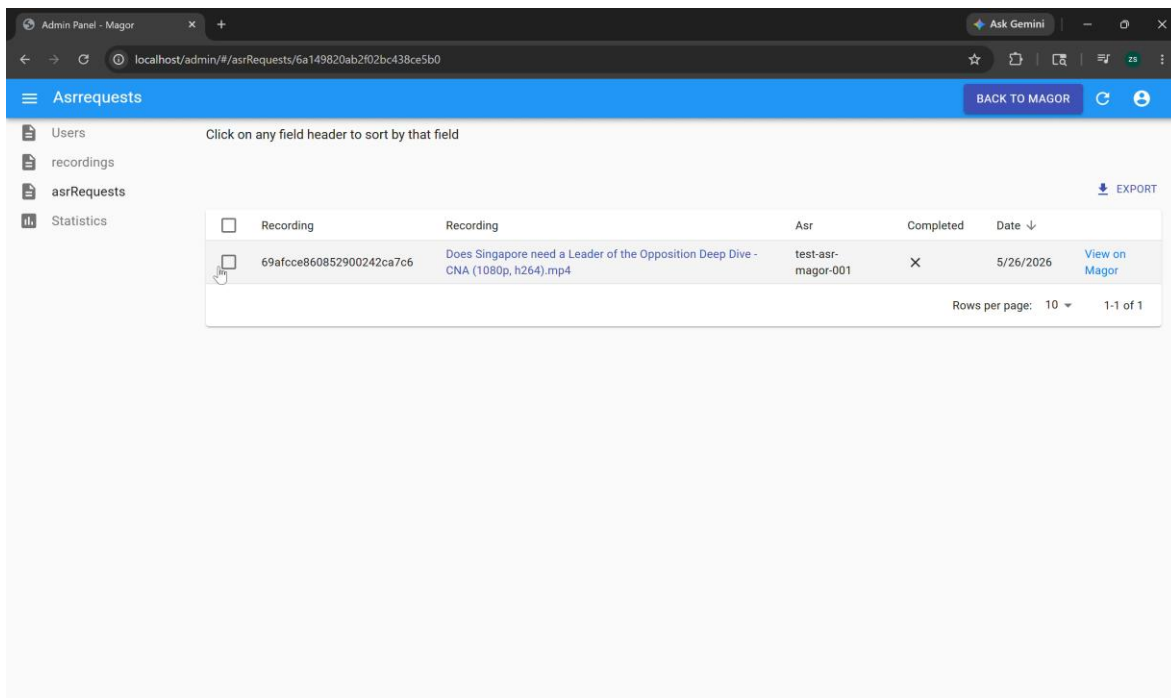
ASR Requests							BACK TO MAGOR		
Users	Click on any field header to sort by that field								
recordings									
ASR Requests									EXPORT
Statistics									
☐	Recording ID	Recording	ASR ID	Status ↓	Date				
☐	69afce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	Success	3/10/2026		View on Magor		
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	Success	3/11/2026		View on Magor		
☐	69b2730fdb3f0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	Success	3/12/2026		View on Magor		
☐	6a14889a49b93b00234a46cd	Original_recording5.mp4	6a1488a263309cdb335cf350	Success	5/26/2026		View on Magor		
☐	6a152d6714f3860023a87170	Bases Juiced, One Out.mp4	6a152d7363309cdb335d4226	Pending	5/26/2026		View on Magor		
							Rows per page: 10	1-5 of 5	

Color coded status display



Task 3 (Fix ASR Request Deletion)

- ASR Deletion error demo:



The screenshot shows a web application interface for 'Admin Panel - Magor'. The browser address bar indicates the URL is `localhost/admin/#/asrRequests/6a149820ab2f02bc438ce5b0`. The page title is 'Asrrequests'. A sidebar on the left contains links for 'Users', 'recordings', 'asrRequests', and 'Statistics'. The main content area displays a table of ASR requests. The table has columns for 'Recording', 'Recording', 'Asr', 'Completed', and 'Date'. A single row is visible with the following data: '69afcce860852900242ca7c6', 'Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4', 'test-asr-magor-001', 'X', and '5/26/2026'. A 'View on Magor' link is present next to the date. At the bottom right of the table, it says 'Rows per page: 10' and '1-1 of 1'. A 'BACK TO MAGOR' button is located in the top right corner of the page.

Recording	Recording	Asr	Completed	Date ↓
69afcce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	test-asr-magor-001	X	5/26/2026



Task 3 (Fix ASR Request Deletion)

- ASR Deletion error code fix:

```
delete: (resource, params) =>
  httpClient(`${apiUrl}/${resource}/${params.id}`, {
    method: 'DELETE',
  }).then(({ json }) => ({ data: json })),
//delete many not supported
deleteMany: (resource, params) => {
  const query = {
    filter: JSON.stringify({ id: params.ids }),
  }
  return httpClient(`${apiUrl}/${resource}?${stringify(query)}`, {
    method: 'DELETE',
    body: JSON.stringify(params.data),
  }).then(({ json }) => ({ data: json }))
  }).then(() => ({ data: params.ids }))
},
```

DataProvider.js
(Frontend)

Delete
API
Request



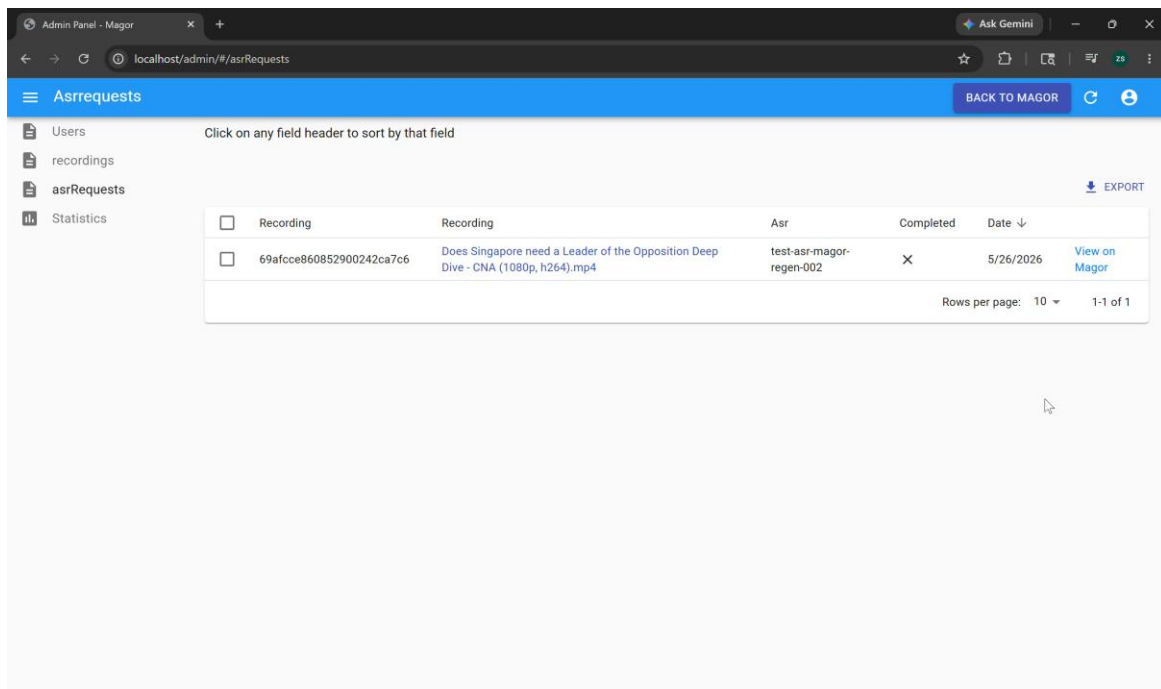
```
79 + const deleteMany = async (req, res) => {
80 +   try {
81 +     const query = await db.checkQueryString(req.query);
82 +     const ids = query.id || query._id || [];
83 +     const deleteQuery = Array.isArray(ids) ? { _id: { $in: ids
84 +       } } : { _id: ids };
85 +     const result = await model.deleteMany(deleteQuery);
86 +     res.status(200).json({ msg: 'DELETED', deletedCount:
87 +       result.deletedCount });
88 +   } catch (error) {
89 +     utils.handleError(res, error);
90 +   }
91 + };
92 + const deleteItem = async (req, res) => {
93 +   try {
94 +     const item = await db.deleteItem(req.params.id, model);
95 +     res.status(200).json(item);
96 +   } catch (error) {
97 +     utils.handleError(res, error);
98 +   }
99 + };
```

transcriptionRequest.js
(Backend)



Task 3 (Fix ASR Request Deletion)

- Complete ASR Deletion demo:



The screenshot shows a web application interface for 'Admin Panel - Magor'. The browser address bar indicates the URL is 'localhost/admin/#/asrRequests'. The page has a blue header bar with the title 'Asrrequests' and a 'BACK TO MAGOR' button. On the left, there is a sidebar menu with options: 'Users', 'recordings', 'asrRequests' (selected), and 'Statistics'. The main content area displays a table of ASR requests. Above the table, there is a note: 'Click on any field header to sort by that field'. The table has columns: 'Recording' (with a checkbox), 'Recording' (text), 'Asr' (text), 'Completed' (text), and 'Date' (with a dropdown arrow). There is also an 'EXPORT' button with a download icon. The table contains one row with the following data: a checkbox, the ID '69afcce860852900242ca7c6', the text 'Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4', the text 'test-asr-magor-regen-002', and the date '5/26/2026'. A link 'View on Magor' is present next to the date. At the bottom right of the table, it says 'Rows per page: 10' and '1-1 of 1'.

☐	Recording	Recording	Asr	Completed	Date ↓	
☐	69afcce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	test-asr-magor-regen-002	X	5/26/2026	View on Magor

Rows per page: 10 1-1 of 1



Task 3 (Fix ASR Request Deletion)

- Past ASR requests that have their recordings deleted are still displayed

The screenshot shows the 'ASR Requests' management page. The interface includes a sidebar with navigation links for Users, recordings, ASR Requests, and Statistics. The main content area displays a table of ASR requests. Annotations highlight specific issues:

- Missing Recording:** A red oval highlights the 'Recording' column for the first row, which is empty.
- Dead Link Redirect:** A red arrow points to the 'View on Magor' link in the first row.
- Status Column:** A red box highlights the 'Status' column, which contains buttons labeled 'Success', 'Pending', and 'Success'.
- Use status to indicate deletion?:** A red arrow points to the 'Status' column with this text.

Additional annotations include 'Missing Recording' pointing to the empty recording cell and 'Dead Link Redirect' pointing to the 'View on Magor' link.

☐	Recording ID	Recording	ASR ID	Status	Date	
☐	6a1530601048bd0023103aa5		6a15306663309cdb335d43e0	Success	5/26/2026	View on Magor
☐	6a152d6714f3860023a87170	Bases Juiced, One Out.mp4	6a152d7363309cdb335d4226	Pending	5/26/2026	View on Magor
☐	69b2730fdb3f0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	Success	3/12/2026	View on Magor
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	Success	3/11/2026	View on Magor
☐	69afcce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	Success	3/10/2026	View on Magor

Rows per page: 10 1-5 of 5

Task 3 (Fix ASR Request Deletion)

- Request status reflects deleted recordings

ASR Requests

BACK TO MAGOR

Users

recordings

ASR Requests

Statistics

Click on any field header to sort by that field

EXPORT

☐	Recording ID	Recording	ASR ID	Status	Date ↓	
☐	6a153848ea5e4a0022d9c2cc		6a15384e63309cdb335d48d8	Deleted	5/26/2026	View on Magor
☐	6a152d6714f3860023a87170	Bases Juiced, One Out.mp4	6a152d7363309cdb335d4226	Pending	5/26/2026	View on Magor
☐	69b2730fdbef0002339be00	YouTube_Parliament-passes-motion-considering-Pri_Media_67m11XNfN3w_001_1080p	69b27321b356c587a85aa827	Success	3/12/2026	View on Magor
☐	69b0b7d6f7044e0023c57d96	Indonesian stocks plunge more than 7 after MSCI flags risks - CNA (144p, h264, youtube).mp4	69b0bb1eb356c587a859c8a4	Success	3/11/2026	View on Magor
☐	69afcce860852900242ca7c6	Does Singapore need a Leader of the Opposition Deep Dive - CNA (1080p, h264).mp4	69afcd4ab356c587a8592411	Success	3/10/2026	View on Magor

Rows per page: 10 1-5 of 5

Plans for Next Week

- **Restore Admin Panel**
 - Automate deletion of ASR requests linked to deleted recordings
 - Modify deletes from recording page to also delete associated files
 - Restore Statistics Page
- **Magor Studio**
 - Update transcription job submission to include speaker count

